

# On chemical flooding models: Riemann problem solutions and viscous fingering

Yulia Petrova

IMPA



<https://yulia-petrova.github.io/>

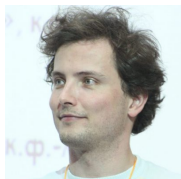


27 July 2022

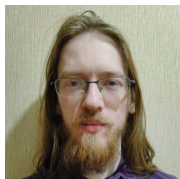
UFRJ - Seminário Luiz Adauto Medeiros de Análise/EDP

## Joint work with

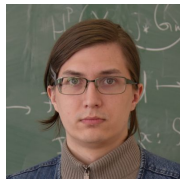
Chebyshev Laboratory, St Petersburg State University, Russia + Texas A&M, USA



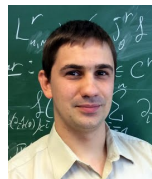
Fedor  
Bakharev



Aleksandr  
Enin



Nikita  
Rastegaev



Sergey  
Tikhomirov



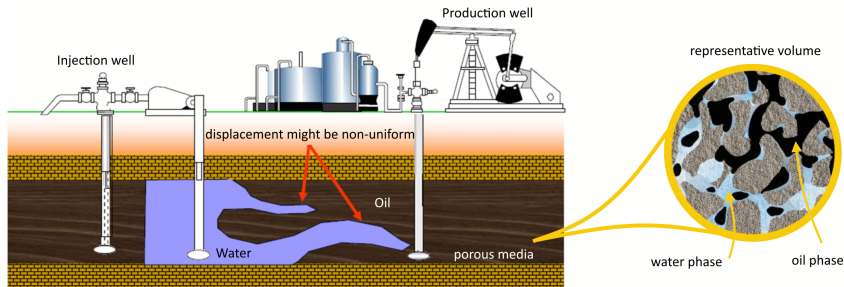
Yalchin  
Efendiev

- Collaboration with Russian petroleum company GazpromNeft (2018–2021)
- The talk is based on:
  - F. Bakharev, A. Enin, Yu. Petrova, N. Rastegaev “Impact of dissipation ratio on vanishing viscosity solutions of the Riemann problem for chemical flooding model”  
See [arXiv:2111.15001](https://arxiv.org/abs/2111.15001)
  - Ongoing research with S. Tikhomirov and Ya. Efendiev “A cascade of two travelling waves in a two-tube model of viscous and gravitational fingering”

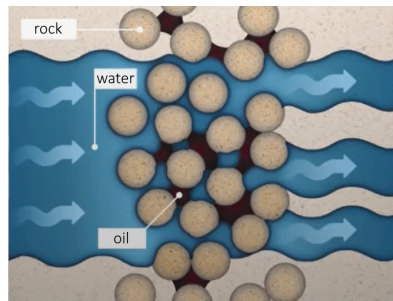
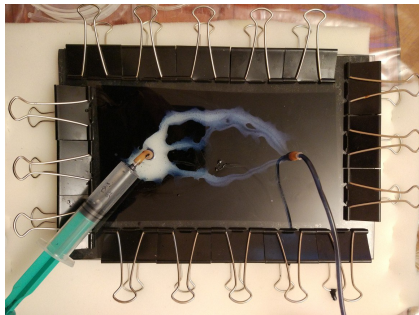
# Motivation

We are interested in the mathematical model of oil recovery. Some features:

- Porous media (averaged models of flow)
- Unknown variables:  $s(t, x)$  — the averaged water saturation in small volume  
 $1 - s$  — the average oil saturation in representative volume
- Relatively small speeds ( $\approx 1$  meter per day): Navier-Stokes  $\rightarrow$  Darcy's law
- Multiphase flow: oil, water, gas.
- Applications to EOR (enhanced oil recovery) methods: chemical, thermal, gas etc



# Problems: macroscopic and microscopic sweep efficiency



- happens due to very viscous oil or inhomogeneous media
- local entrapment of oil in pores due to high capillary pressure

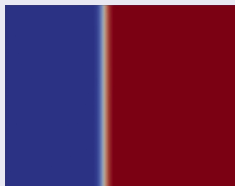
## Possible solution

- Inject gas ( $\text{CO}_2$ , natural) to decrease the oil viscosity
- Add **chemicals (polymer)** to increase the water viscosity
- Add **chemicals (surfactant)** that reduce the surface tension etc

# Fundamental research: two main directions

## 1-dim in spatial variable

- Stable displacement



- main questions: find an exact solution to a Riemann problem; analyze stability
- hyperbolic conservation laws

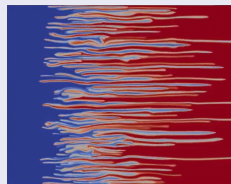
$$s_t + f(s, c)_x = 0,$$

$$(cs + a(c))_t + (cf(s, c))_x = 0.$$

Example: chemical flooding model

## 2-dim (or 3-dim) in spatial variable

- Unstable displacement



- source of instability: water and oil/polymer have different viscosities
- viscous fingering phenomenon

$$c_t + u \cdot \nabla c = \varepsilon \Delta c,$$

$$\operatorname{div}(u) = 0,$$

$$u = -\nabla p / \mu(c).$$

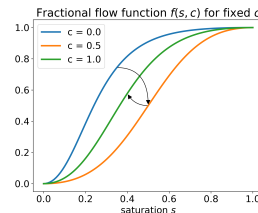
Example: Peaceman model

# Problem statement

Chemical flooding can be described as the system of conservation laws ( $x \in \mathbb{R}, t > 0$ ):

$$\begin{aligned} s_t + f(s, c)_x &= 0, & (\text{conservation of water}) \\ (cs + a(c))_t + (cf(s, c))_x &= 0. & (\text{conservation of chemical}) \end{aligned} \quad (1)$$

- $s = s(x, t)$  — water phase saturation;
- $f(s, c)$  — fractional flow function (usually  $S$ -shaped);
- $c = c(x, t)$  — concentration of a chemical in water;
- $a(c)$  — adsorption of a chemical on a rock (usually increasing, concave).



Initial data:

$$(s, c)|_{t=0} = \begin{cases} (s^L, c^L), & \text{if } x \leq 0, \\ (s^R, c^R), & \text{if } x > 0, \end{cases} \quad (2)$$

Aim:

Find a solution to initial-value problem (1)–(2) when  $f$  depends non-monotonically on  $c$ .

NB: Non-monotone dependence appears in surfactant flooding, low salinity water flooding etc

## Preview of results

- weak solutions  $\Rightarrow$  non-uniqueness of solutions to a Riemann problem

## Preview of results

- weak solutions  $\Rightarrow$  non-uniqueness of solutions to a Riemann problem
- use vanishing viscosity criterion — add small diffusion/capillary terms

$$\begin{aligned}
 s_t + f(s, c)_x &= \varepsilon_c (A(s, c) s_x)_x, \\
 (cs + a(c))_t + (cf(s, c))_x &= \varepsilon_c (cA(s, c) s_x)_x + \varepsilon_d c_{xx}.
 \end{aligned}$$

## Preview of results

- weak solutions  $\Rightarrow$  non-uniqueness of solutions to a Riemann problem
- use vanishing viscosity criterion — add small diffusion/capillary terms

$$\begin{aligned}
 s_t + f(s, c)_x &= \varepsilon_c (A(s, c) s_x)_x, \\
 (cs + a(c))_t + (cf(s, c))_x &= \varepsilon_c (cA(s, c) s_x)_x + \varepsilon_d c_{xx}.
 \end{aligned}$$

- $f(s, c)$  monotone in  $c \Rightarrow$  uniqueness of vanishing viscosity solution (1988)

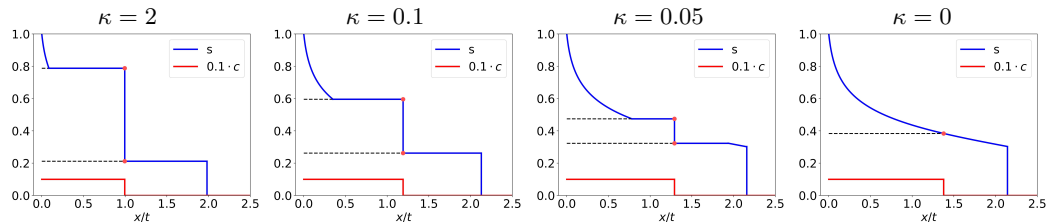
# Preview of results

- weak solutions  $\Rightarrow$  non-uniqueness of solutions to a Riemann problem
- use vanishing viscosity criterion — add small diffusion/capillary terms

$$s_t + f(s, c)_x = \varepsilon_c (A(s, c) s_x)_x,$$

$$(cs + a(c))_t + (cf(s, c))_x = \varepsilon_c (cA(s, c) s_x)_x + \varepsilon_d c_{xx}.$$

- $f(s, c)$  monotone in  $c \Rightarrow$  uniqueness of vanishing viscosity solution (1988)
- Main idea of the 1st part of the talk:  $f(s, c)$  non-monotone in  $c \Rightarrow$  exist multiple vanishing viscosity solutions, depending on ratio  $\kappa = \varepsilon_d / \varepsilon_c$



NB: the appeared shock is known as undercompressive (transitional).

# Hyperbolic systems of conservation laws\*

$$G(u)_t + F(u)_x = 0 \quad (3)$$

- $G(u)$  — **accumulation function** (conserved quantities)
- $F(u)$  — **flux function** (flux of conserved quantities)

Simplest example: wave equation

$$y_{tt} - c^2 y_{xx} = 0 \quad (\text{J. d'Alembert, 1750})$$

can be rewritten as a system of two first-order equations on the state-vector  $u = \begin{pmatrix} y_x \\ y_t \end{pmatrix}$

$$u_t + Du_x = 0, \quad \text{with} \quad D = \begin{pmatrix} 0 & -1 \\ -c^2 & 0 \end{pmatrix}$$

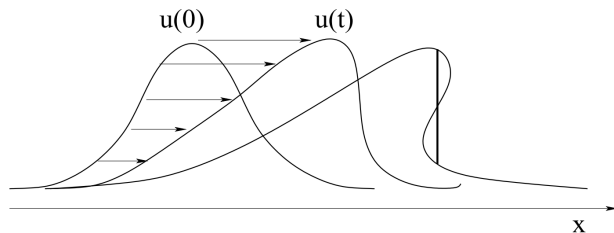
- eigenvalues  $\lambda_1 = c$  and  $\lambda_2 = -c$  are **real**, the system is **hyperbolic**. Solution contain two **wave modes** that propagate at the **velocities**  $\lambda_1$  and  $\lambda_2$ .

\* For more details see e.g. "Hyperbolic conservation laws: an illustrated tutorial" by Alberto Bressan.

# Hyperbolic systems of conservation laws

$$u_t + \left( \frac{u^2}{2} \right)_x = 0 \quad (\text{Burger's equation, 1948})$$

- non-linearity implies wave speed  $\lambda(u) = u$  depends on state  $u$
- So the wave can spread (rarefaction wave) or concentrate (shock wave)



$$u_t + (f(u))_x = 0 \quad (\text{Buckley-Leverett equation})$$

- existence, uniqueness was established by Olga Oleinik (1957)

# Riemann problem (1858)

- Riemann solved the initial-value problem with data having a **single jump**

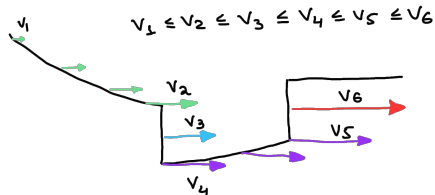
$$u|_{t=0} = \begin{cases} u^L, & x \leq 0; \\ u^R, & x > 0. \end{cases}$$

- took advantage of the **scale invariance** of the equations and the data:

$$u(\alpha x, \alpha t) = u(x, t) \quad \text{for all } \alpha > 0$$

- solution to a Riemann problem is important because:
  - it appears in a long-term behavior of Cauchy problem
  - helps to prove the existence of solutions to Cauchy problem (Glimm's method)
  - helps to construct numerical solution (Godunov method)

Any solution to a Riemann problem consists of a sequence of rarefaction or shock waves (and constant states) that are **compatible by speeds**

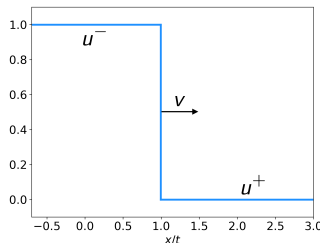


# Shock waves: RH condition and admissibility criteria

- discontinuous solutions are defined in the sense of distributions (**weak form**)
- for a shock wave from  $u^-$  to  $u^+$  moving with velocity  $v$ , the weak condition amounts to the following **Rankine-Hugoniot** condition (RH)

$$-v G(u^-) + F(u^-) = -v G(u^+) + F(u^+) \quad (\text{RH})$$

- RH means conservation: what flows into left side flows out of the right side



- exist various admissibility criteria: entropy, Lax, Liu, **vanishing viscosity**

# Travelling wave solutions of diffusive system (Hopf, 1948)

- **Vanishing viscosity criteria:** consider a diffusive system of conservation laws

$$G(u)_t + F(u)_x = \varepsilon [B(u) u_x]_x, \quad \varepsilon \rightarrow 0$$

# Travelling wave solutions of diffusive system (Hopf, 1948)

- **Vanishing viscosity criteria:** consider a diffusive system of conservation laws

$$G(u)_t + F(u)_x = \varepsilon [B(u) u_x]_x, \quad \varepsilon \rightarrow 0$$

- $u(x, t) = \hat{u}(\xi)$  with  $\xi := x - v t$  for a fixed shock velocity  $v$
- reduction to first-order system of ordinary differential equations:

$$\varepsilon B(\hat{u}) \hat{u}_\xi = -v [G(\hat{u}) - G(u^-)] + F(\hat{u}) - F(u^-)$$

# Travelling wave solutions of diffusive system (Hopf, 1948)

- **Vanishing viscosity criteria:** consider a diffusive system of conservation laws

$$G(u)_t + F(u)_x = \varepsilon [B(u) u_x]_x, \quad \varepsilon \rightarrow 0$$

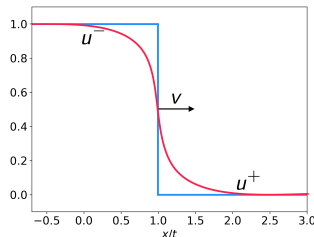
- $u(x, t) = \hat{u}(\xi)$  with  $\xi := x - v t$  for a fixed shock velocity  $v$
- reduction to first-order system of ordinary differential equations:

$$\varepsilon B(\hat{u}) \hat{u}_\xi = -v [G(\hat{u}) - G(u^-)] + F(\hat{u}) - F(u^-)$$

- $u^-$  and  $u^+$  are fixed points and we look for an orbit connecting them

$$\hat{u}(-\infty) = u^-, \quad \hat{u}(+\infty) = u^+$$

- travelling wave solution approaches the jump discontinuity in  $L^1$  as  $\varepsilon \rightarrow 0^+$



## Historical review: zero adsorption

$$\begin{aligned}
 s_t + f(s, c)_x &= 0, \\
 (sc)_t + (cf(s, c))_x &= 0.
 \end{aligned}
 \quad (s, c)|_{t=0} = \begin{cases} (s_L, c_L), & \text{if } x \leq 0, \\ (s_R, c_R), & \text{if } x > 0, \end{cases} \quad (4)$$

## Historical review: zero adsorption

$$\begin{aligned}
 s_t + f(s, c)_x &= 0, \\
 (sc)_t + (cf(s, c))_x &= 0.
 \end{aligned}
 \quad (s, c)|_{t=0} = \begin{cases} (s_L, c_L), & \text{if } x \leq 0, \\ (s_R, c_R), & \text{if } x > 0, \end{cases} \quad (4)$$

Can be rewritten in a more symmetric form:

$$\begin{aligned}
 s_t + (sg(s, b))_x &= 0, & b = sc & \text{— total amount of chemicals} \\
 b_t + (bg(s, b))_x &= 0. & g = f/s & \text{— new flux function}
 \end{aligned}$$

- 1980 — elasticity theory (Keyfitz, Kranzer):  $g(s, b) = \tilde{g}(s^2 + b^2)$ .

## Historical review: zero adsorption

$$\begin{aligned}
 s_t + f(s, c)_x &= 0, \\
 (sc)_t + (cf(s, c))_x &= 0.
 \end{aligned}
 \quad (s, c)|_{t=0} = \begin{cases} (s_L, c_L), & \text{if } x \leq 0, \\ (s_R, c_R), & \text{if } x > 0, \end{cases} \quad (4)$$

Can be rewritten in a more symmetric form:

$$\begin{aligned}
 s_t + (sg(s, b))_x &= 0, & b = sc & \text{— total amount of chemicals} \\
 b_t + (bg(s, b))_x &= 0. & g = f/s & \text{— new flux function}
 \end{aligned}$$

- 1980 — elasticity theory (Keyfitz, Kranzer):  $g(s, b) = \tilde{g}(s^2 + b^2)$ .
- 1980 — polymer flooding (Isaacson, Glimm, Temple):  $f(s, c)$  monotone in  $c$

## Historical review: zero adsorption

$$\begin{aligned}
 s_t + f(s, c)_x &= 0, \\
 (sc)_t + (cf(s, c))_x &= 0.
 \end{aligned}
 \quad (s, c)|_{t=0} = \begin{cases} (s_L, c_L), & \text{if } x \leq 0, \\ (s_R, c_R), & \text{if } x > 0, \end{cases} \quad (4)$$

Can be rewritten in a more symmetric form:

$$\begin{aligned}
 s_t + (sg(s, b))_x &= 0, & b = sc & \text{— total amount of chemicals} \\
 b_t + (bg(s, b))_x &= 0. & g = f/s & \text{— new flux function}
 \end{aligned}$$

- 1980 — elasticity theory (Keyfitz, Kranzer):  $g(s, b) = \tilde{g}(s^2 + b^2)$ .
- 1980 — polymer flooding (Isaacson, Glimm, Temple):  $f(s, c)$  monotone in  $c$

Interesting properties:

- has coordinate system of Riemann invariants
- rarefaction and shock curves coincide

Problems:

- **non-strictly** hyperbolic
- has **contact discontinuities**

## Historical review: zero adsorption

$$\begin{aligned}
 s_t + f(s, c)_x &= 0, \\
 (sc)_t + (cf(s, c))_x &= 0.
 \end{aligned}
 \quad (s, c)|_{t=0} = \begin{cases} (s_L, c_L), & \text{if } x \leq 0, \\ (s_R, c_R), & \text{if } x > 0, \end{cases} \quad (4)$$

Can be rewritten in a more symmetric form:

$$\begin{aligned}
 s_t + (sg(s, b))_x &= 0, & b = sc & \text{— total amount of chemicals} \\
 b_t + (bg(s, b))_x &= 0. & g = f/s & \text{— new flux function}
 \end{aligned}$$

- 1980 — elasticity theory (Keyfitz, Kranzer):  $g(s, b) = \tilde{g}(s^2 + b^2)$ .
- 1980 — polymer flooding (Isaacson, Glimm, Temple):  $f(s, c)$  monotone in  $c$

Interesting properties:

- has coordinate system of Riemann invariants
- rarefaction and shock curves coincide

Problems:

- **non-strictly** hyperbolic
- has **contact discontinuities**

Work in progress with D. Marchesin, B. Plohr: justification of Isaacson-Glimm admissibility criterion & construction of a solution for the case when  $f(s, c)$  is non-monotone in  $c$

## Historical review: non-zero adsorption

$$\begin{aligned}
 s_t + f(s, c)_x &= 0, \\
 (sc + a(c))_t + (cf(s, c))_x &= 0.
 \end{aligned}
 \quad (s, c)|_{t=0} = \begin{cases} (s_L, c_L), & \text{if } x \leq 0, \\ (s_R, c_R), & \text{if } x > 0, \end{cases} \quad (5)$$

## Historical review: non-zero adsorption

$$\begin{aligned}
 s_t + f(s, c)_x &= 0, \\
 (sc + a(c))_t + (cf(s, c))_x &= 0.
 \end{aligned}
 \quad (s, c)|_{t=0} = \begin{cases} (s_L, c_L), & \text{if } x \leq 0, \\ (s_R, c_R), & \text{if } x > 0, \end{cases} \quad (5)$$

Historical review:

- 1988 — JW,  $f(s, c)$  monotone in  $c$ . Found a unique vanishing viscosity solution.

When  $f(s, c)$  **non-monotone** in  $c$ , **multiple** vanishing viscosity **solutions are possible**.

- 2017 — W. Shen, gave some examples in Lagrangian coordinates. See also
- 1986 — Entov, Kerimov, non-rigorous consideration of the non-monotone case.

# Historical review: non-zero adsorption

$$\begin{aligned} s_t + f(s, c)_x &= 0, \\ (sc + a(c))_t + (cf(s, c))_x &= 0. \end{aligned} \quad (s, c)|_{t=0} = \begin{cases} (s_L, c_L), & \text{if } x \leq 0, \\ (s_R, c_R), & \text{if } x > 0, \end{cases} \quad (5)$$

Historical review:

- 1988 — JW,  $f(s, c)$  monotone in  $c$ . Found a unique vanishing viscosity solution.

When  $f(s, c)$  **non-monotone** in  $c$ , **multiple** vanishing viscosity **solutions are possible**.

- 2017 — W. Shen, gave some examples in Lagrangian coordinates. See also
- 1986 — Entov, Kerimov, non-rigorous consideration of the non-monotone case.

## Proposition (Johansen-Winther, 1988 (JW))

*The solution  $c(x, t)$  is monotone in  $x$  for any  $t > 0$ . If  $a(c)$  is concave, then*

- *is constant if  $c_L = c_R$ ;*
- *contains several rarefaction waves if  $c_L < c_R$ ;*
- *contains exactly one shock wave if  $c_L > c_R$ :*

$$(1, 1) \xrightarrow{c=1} u^- \xrightarrow{c\text{-shock}} u^+ \xrightarrow{c=0} (0, 0)$$

# Dissipative system

To define a shock wave between  $u^-$  and  $u^+$  we consider dissipative system:

$$\begin{aligned} s_t + f(s, c)_x &= \varepsilon_c (A(s, c) s_x)_x, \\ (cs + \alpha)_t + (cf(s, c))_x &= \varepsilon_c (cA(s, c) s_x)_x + \varepsilon_d c_{xx}, \\ \alpha_t &= \varepsilon_r^{-1} (a(c) - \alpha). \end{aligned}$$

- $\varepsilon_c$  — dimensionless capillary pressure
- $\varepsilon_d$  — dimensionless diffusion term
- $\varepsilon_r$  — dimensionless relaxation time
- $A(s, c)$  — capillary pressure term
- $\alpha = \alpha(x, t)$  — dynamic adsorption

We consider two particular cases:

Capillarity and Diffusion

$$\begin{aligned} s_t + f(s, c)_x &= \varepsilon_c (A(s, c) s_x)_x, \\ (cs + a(c))_t + (cf(s, c))_x &= \varepsilon_c (cA(s, c) s_x)_x + \varepsilon_d c_{xx}, \end{aligned}$$

Capillarity and Dynamic Adsorption

$$\begin{aligned} s_t + f(s, c)_x &= \varepsilon_c (A(s, c) s_x)_x, \\ (cs + \alpha)_t + (cf(s, c))_x &= \varepsilon_c (cA(s, c) s_x)_x, \\ \alpha_t &= \varepsilon_r^{-1} (a(c) - \alpha). \end{aligned}$$

\* Bedrikovetsky, Pavel. Mathematical theory of oil and gas recovery: with applications to ex-USSR oil and gas fields. Vol. 4. Springer Science & Business Media, 1993.

# Vanishing viscosity admissibility criteria

## Admissibility criteria

A shock between  $u^+ = (s^+, c^+)$  and  $u^- = (s^-, c^-)$  is admissible if it could be obtained as a limit of smooth travelling wave solutions of the dissipative system as  $\varepsilon_{c,d} \rightarrow 0$

$$\begin{aligned} s_t + f(s, c)_x &= \varepsilon_c (A(s, c) s_x)_x, \\ (cs + a(c))_t + (cf(s, c))_x &= \varepsilon_c (cA(s, c) s_x)_x + \varepsilon_d c_{xx}. \end{aligned} \tag{6}$$

# Vanishing viscosity admissibility criteria

## Admissibility criteria

A shock between  $u^+ = (s^+, c^+)$  and  $u^- = (s^-, c^-)$  is admissible if it could be obtained as a limit of smooth travelling wave solutions of the dissipative system as  $\varepsilon_{c,d} \rightarrow 0$

$$\begin{aligned} s_t + f(s, c)_x &= \varepsilon_c (A(s, c) s_x)_x, \\ (cs + a(c))_t + (cf(s, c))_x &= \varepsilon_c (cA(s, c) s_x)_x + \varepsilon_d c_{xx}. \end{aligned} \tag{6}$$

Searching for travelling wave solutions  $s = s(\xi)$ ,  $c = c(\xi)$ ,  $\xi := \varepsilon_c^{-1}(x - vt)$  with boundary conditions  $s(\pm\infty) = s^\pm$ ,  $c(-\infty) = 1$ ,  $c(+\infty) = 0$ , we arrive at

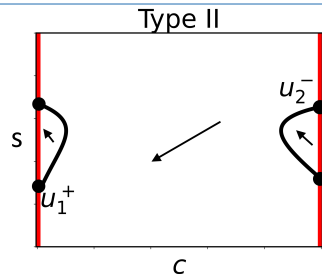
$$\begin{aligned} s_\xi &= A^{-1}(s, c)(f(s, c) - v(s + d_1)), \\ c_\xi &= v\kappa^{-1}(d_1 c - d_2 - a(c)). \end{aligned} \tag{7}$$

- Parameters:  $\kappa = \varepsilon_d/\varepsilon_c$  and  $v$ .  $d_1 = a(1)$  and  $d_2 = 0$  — some constants.
- We are interested for which values of  $(v, \kappa)$  there exist orbits connecting two saddle points  $u^\pm$  (or saddle-nodes) due to compatibility of speeds condition

# Main result

Consider a dynamical system under technical restrictions on functions  $f$ ,  $a$ ,  $A$ :

$$\begin{aligned} s_\xi &= A^{-1}(s, c)(f(s, c) - v(s + d_1)), \\ c_\xi &= vk^{-1}(d_1c - d_2 - a(c)). \end{aligned}$$



**Theorem (Bakharev, Enin, P., Rastegaev, 2021, arxiv:2111.15001)**

*There exist  $0 < v_{\min} < v_{\max} < \infty$ , such that for every  $\kappa = \varepsilon_d / \varepsilon_c \in (0, +\infty)$ , there exist unique*

- *points  $s^-(\kappa) \in [0, 1]$  and  $s^+(\kappa) \in [0, 1]$ ;*
- *velocity  $v(\kappa) \in [v_{\min}, v_{\max}]$ ,*

*such that there exists a travelling wave, connecting two saddle points  $u^-(\kappa) = (s^-(\kappa), 1)$  and  $u^+(\kappa) = (s^+(\kappa), 0)$  with velocity  $v(\kappa)$ . Moreover,  $v(\kappa)$  is monotone and continuous;  $v(\kappa) \rightarrow v_{\min}$  as  $\kappa \rightarrow \infty$ ;  $v(\kappa) \rightarrow v_{\max}$  as  $\kappa \rightarrow 0$ .*

# Solution construction algorithm

1. From  $\kappa$  we calculate  $v(\kappa)$  (binary search).
2. From  $v$  we determine  $s^-(v)$  and  $s^+(v)$  via Rankine-Hugoniot condition.
3. Construct waves  $(1, 1) \rightarrow (s^-(v), 1)$  and  $(s(v), 0) \rightarrow (0, 0)$ .

Example: “boomerang” model:

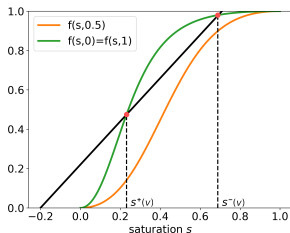


Figure 1: Flux functions

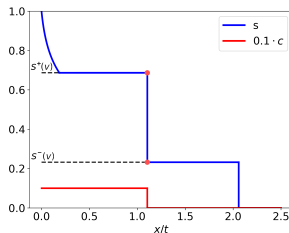


Figure 2: Solution  $s$  and  $c$

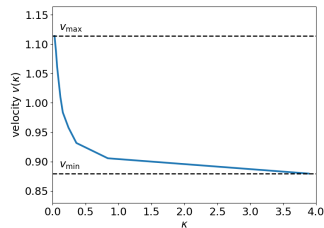


Figure 3: Function  $v(\kappa)$

# Possible future directions

- Existence and uniqueness:
  - to any Riemann problem (in progress)  
NB: appear transitional rarefaction waves
  - to a Cauchy problem  
NB: standard techniques (Glimm scheme and Temple functional) do not work due to non-linear resonance
- Stability and convergence
  - asymptotic stability for the travelling waves solutions?
  - does the whole solution of the dissipative system converge to a solution of non-dissipative system?
- Generalizations:
  - connection between the zero and non-zero adsorption models (in progress with D. Marchesin and B. Plohr)
  - three-phase flow with chemicals (water with chemicals, oil and gas):  
NB: travelling wave dynamical system will become three-dimensional, thus the analysis will be more complex.

# Possible future directions

- Existence and uniqueness:
  - to any Riemann problem (in progress)  
NB: appear transitional rarefaction waves
  - to a Cauchy problem  
NB: standard techniques (Glimm scheme and Temple functional) do not work due to non-linear resonance
- Stability and convergence
  - asymptotic stability for the travelling waves solutions?
  - does the whole solution of the dissipative system converge to a solution of non-dissipative system?
- Generalizations:
  - connection between the zero and non-zero adsorption models (in progress with D. Marchesin and B. Plohr)
  - three-phase flow with chemicals (water with chemicals, oil and gas):  
NB: travelling wave dynamical system will become three-dimensional, thus the analysis will be more complex.

Questions?

## 2-dim problems

---

Show video

# System of equations

1. Transport of species ( $\varepsilon \geq 0$ )

$$c_t + u \cdot \nabla c = \varepsilon \Delta c$$

2. Incompressibility condition

$$\operatorname{div}(u) = 0$$

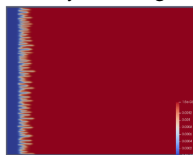
- 3a. Darcy's law (viscosity-driven)

$$u = -k \cdot m(c) \cdot \nabla p$$

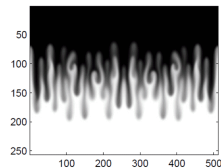
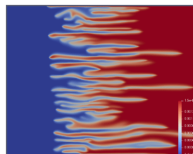
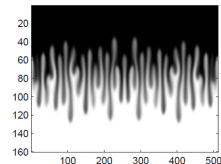
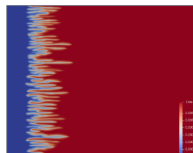
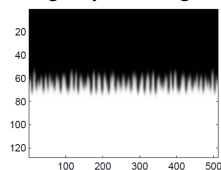
- 3b. Darcy's law (gravity-driven)

$$u = -\nabla p + (0, c)$$

viscosity-driven fingers



gravity-driven fingers



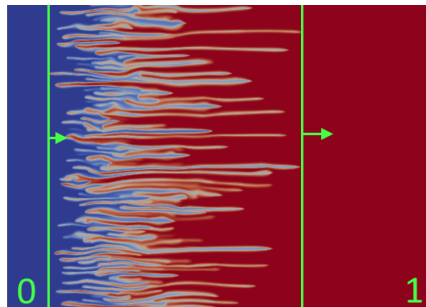
NB: 1 + 2 + 3b for  $\varepsilon = 0$  is known as IPM (incompressible porous media equation)

# Questions of interest

1. Well-posedness:
  - existence of a global solution vs finite-time blow-up:  
active scalar:  $u = A(c)$  — singular integral operator (like in SQG)

## Questions of interest

1. Well-posedness:
  - existence of a global solution vs finite-time blow-up:  
active scalar:  $u = A(c)$  — singular integral operator (like in SQG)
2. Mixing zone:
  - many laboratory and numerical experiments show **linear growth of the mixing zone**



\* Nijjer J., Hewitt D., and Neufeld J. The dynamics of miscible viscous fingering from onset to shutdown. *Journal of Fluid Mechanics* 837 (2018): 520-545.

# Questions of interest

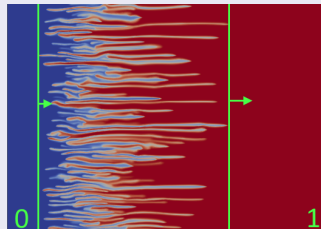
1. Well-posedness:
  - existence of a global solution vs finite-time blow-up:  
active scalar:  $u = A(c)$  — singular integral operator (like in SQG)
2. Mixing zone:
  - many laboratory and numerical experiments show **linear growth of the mixing zone**
  - the only mathematically rigorous result on estimates of speed of the linear growth
    - F. Otto, G. Menon'2006 — for a simplified model of Darcy's law

## Transverse flow equilibrium model (F. Otto, G. Menon'2006)

- get rid of pressure

$$u = -\frac{m(c)}{\text{avg}_y(m(c))}$$

- used techniques of sub / supersolutions
- estimated the mixing zone by two travelling waves connecting 0 to 1

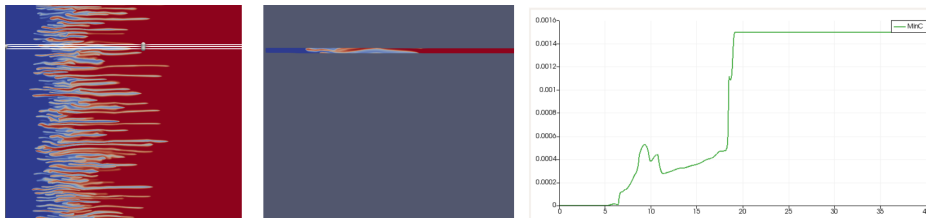


# Questions of interest

1. Well-posedness:
  - existence of a global solution vs finite-time blow-up:  
active scalar:  $u = A(c)$  — singular integral operator (like in SQG)
2. Mixing zone:
  - many laboratory and numerical experiments show **linear growth of the mixing zone**
  - the only mathematically rigorous result on estimates of speed of the linear growth
    - F. Otto, G. Menon'2006 — for a simplified model of Darcy's law

## Our observation from numerical modelling:

There exists some intermediate concentration that makes fingers move slower (for  $\varepsilon > 0$ )



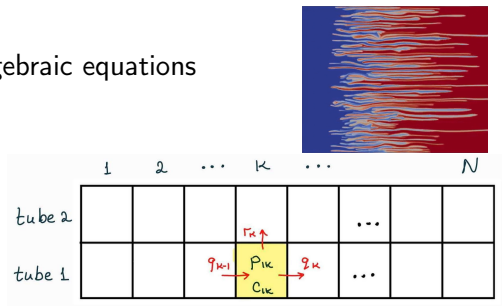
# Toy model of viscous fingering (work in progress)

## Discrete case

- system of  $2N$  ODEs and  $N$  algebraic equations

### Unknowns:

- $c_{1k}(t), c_{2k}(t)$  — concentrations
- $p_{1k}, p_{2k}$  — pressures
- $q_k(t), r_k(t)$  — velocities

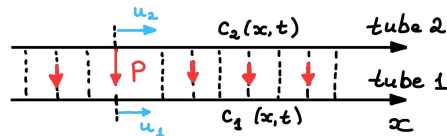


## Continuous case

- two coupled advection-diffusion eqs

$$\partial_t c_1 = -\partial_x(u_1 c_1) + P + \varepsilon \partial_{xx} c_1,$$

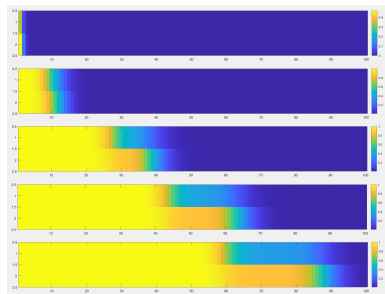
$$\partial_t c_2 = -\partial_x(u_2 c_2) - P + \varepsilon \partial_{xx} c_2.$$



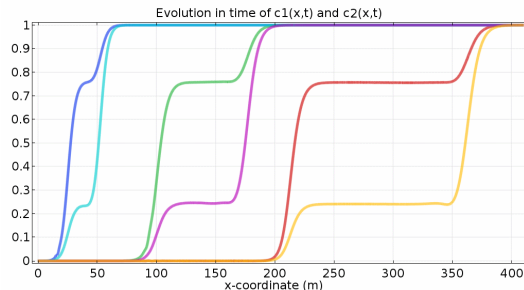
Flow between tubes:  $P = (-1)^{1,2} \partial_x u_{1,2} c_{1,2}$ . Work in progress with S. Tikhomirov, Y. Efendiev.

# “Toy models” of fingering: numerical experiments

Discrete setting

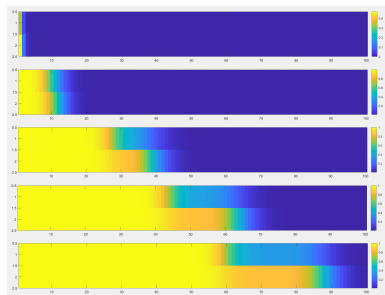


Continuous setting

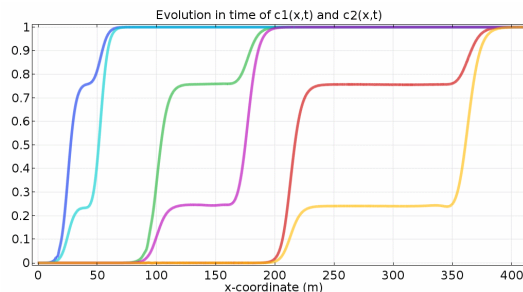


# “Toy models” of fingering: numerical experiments

Discrete setting



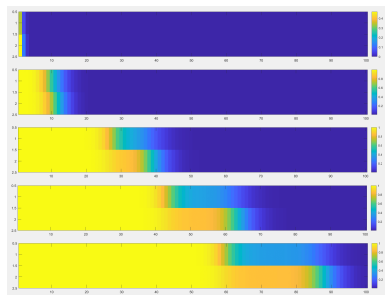
Continuous setting



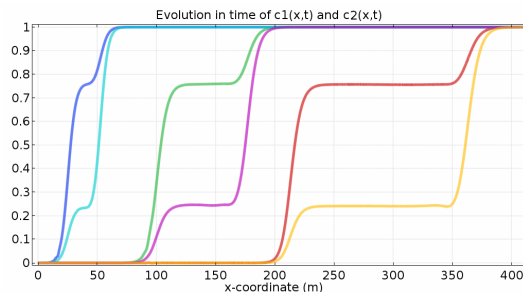
Result of experiments: cascade of two travelling waves:  $(0,0) \xrightarrow{TW_1} (c_1^*, c_2^*) \xrightarrow{TW_2} (1,1)$

# “Toy models” of fingering: numerical experiments

Discrete setting



Continuous setting



Result of experiments: cascade of two travelling waves:  $(0, 0) \xrightarrow{TW_1} (c_1^*, c_2^*) \xrightarrow{TW_2} (1, 1)$

(simplified gravitational fingering)

$$u_1 = 0.5(c_2 - c_1) = -u_2$$

(simplified viscous fingering)

$$u_1 = \frac{2m(c_1)}{m(c_1) + m(c_2)} = 2 - u_2$$

(Darcy's law with pressure)

Biot-Savart Law (non-local)

## “Toy models” of fingering: theoretical approach

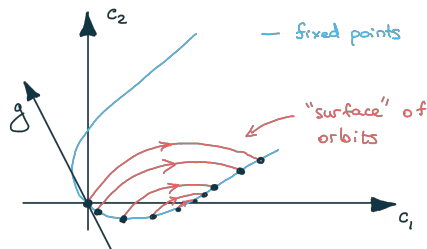
For a travelling wave  $c_1 = c_1(x - vt)$  and  $c_2 = c_2(x - vt)$  we have a dynamical system.

# Travelling wave dynamical system

$$c'_1 = g_1,$$

$$g_1' = (u_1 - v)g_1,$$

$$c'_2 = (u_1 - v)c_1 + (u_2 - v)c_2 - g_1.$$



# “Toy models” of fingering: theoretical approach

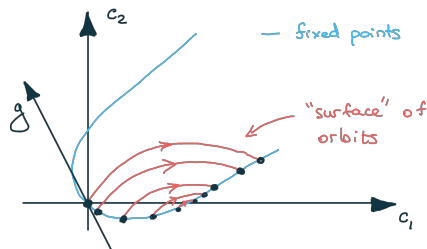
For a travelling wave  $c_1 = c_1(x - vt)$  and  $c_2 = c_2(x - vt)$  we have a dynamical system.

## Travelling wave dynamical system

$$c_1' = g_1,$$

$$g_1' = (u_1 - v)g_1,$$

$$c_2' = (u_1 - v)c_1 + (u_2 - v)c_2 - g_1.$$



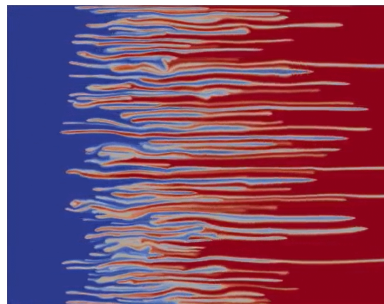
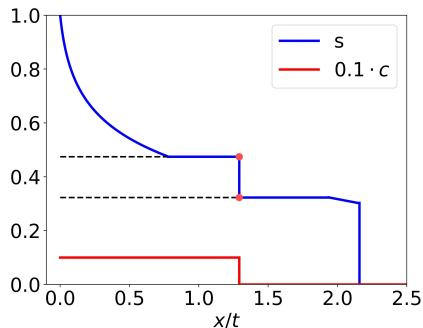
Challenges:

- Prove the existence of a cascade of two travelling waves in 2-tube model under Darcy's law. Is this configuration stable?
- Can one calculate the exact velocities and intermediate concentrations?
- Do these concentrations correspond to ones observed in 2-dim modelling?

# Thank you for your attention!

yulia.petrova@impa.br

<https://yulia-petrova.github.io/>



# Literature:

## Own works:

1. F. Bakharev, A. Enin, Yu. Petrova, N. Rastegaev, Impact of dissipation ratio on vanishing viscosity solutions of the Riemann problem for chemical flooding model. arXiv:2111.15001. Under consideration in a journal.
2. Yu. Petrova, D. Marchesin, B. Plohr, On admissibility criteria for contact discontinuities in Glimm-Isaacson model arising in chemical flooding. Work in progress. See slides.
3. Ya. Efendiev, S. Tikhomirov, Yu. Petrova, A cascade of two travelling waves in a two-tube model of viscous and gravitational fingering. Work in progress.

## Other works:

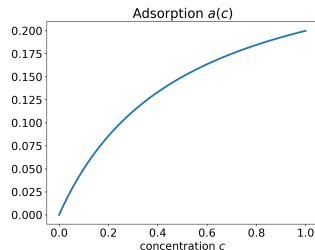
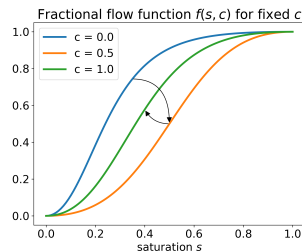
1. Johansen, T. and Winther, R., 1988. The solution of the Riemann problem for a hyperbolic system of conservation laws modeling polymer flooding. SIMA, 19(3), pp.541-566.
2. Shen, W., 2017. On the uniqueness of vanishing viscosity solutions for Riemann problems for polymer flooding. Nonlinear Differential Equations and Applications NoDEA, 24(4), pp.1-25.
3. Entov, V.M. and Kerimov, Z.A., 1986. Displacement of oil by an active solution with a nonmonotonic effect on the flow distribution function. Fluid Dynamics, 21(1), pp.64-70.
4. Keyfitz, B.L. and Kranzer, H.C., 1980. A system of non-strictly hyperbolic conservation laws arising in elasticity theory. Archive for Rational Mechanics and Analysis, 72(3), pp.219-241.

# Literature

5. Isaacson E., 1980. Global solution of a Riemann problem for a non-strictly hyperbolic system of conservation laws arising in enhanced oil recovery // Rockefeller University, NY preprint.
6. Bressan, A., 2013. Hyperbolic conservation laws: an illustrated tutorial. In Modelling and optimisation of flows on networks (pp. 157-245). Springer, Berlin, Heidelberg.
7. Bedrikovetsky, P. Mathematical theory of oil and gas recovery: with applications to ex-USSR oil and gas fields. Vol. 4. Springer Science & Business Media, 1993.
8. Menon, G. and Otto, F., 2006. Diffusive slowdown in miscible viscous fingering. Communications in Mathematical Sciences, 4(1), pp.267-273.
9. Nijjer J., Hewitt D., and Neufeld J. The dynamics of miscible viscous fingering from onset to shutdown. Journal of Fluid Mechanics 837 (2018): 520-545.

# Restrictions on $f$ and $a$

- (F1)  $f \in C^2([0, 1]^2)$ ;  $f(0, c) = 0$ ;  $f(1, c) = 1$ ;
- (F2)  $f_s(s, c) > 0$  for  $s \in (0, 1)$ ,  $c \in [0, 1]$ ;  
 $f_s(0, c) = f_s(1, c) = 0$ ;
- (F3)  $f$  is S-shaped in  $s$ ;
- (F4)  $f$  is **non-monotone** in  $c$ :  
 $\forall s \in (0, 1) \exists c^*(s) \in (0, 1)$ :
- $f_c(s, c) < 0$  for  $0 < s < 1$ ,  $0 < c < c^*(s)$ ;
  - $f_c(s, c) > 0$  for  $0 < s < 1$ ,  $c^*(s) < c < 1$ ;
- (A)  $A$  is bounded from zero and infinity;
- (a)  $a \in C^2$ ,  $a(0) = 0$ ,  $a$  is strictly increasing and concave.



NB: assumptions are just the simplest possible ones. Could be generalised.

# Scheme of proof

The Theorem can be divided into simpler statements:

- $\forall v \in [v_{\min}, v_{\max}] \quad \exists! \kappa(v)$ : there is a saddle-to-saddle travelling wave with  $\kappa(v)$ .
- $\kappa(v)$  is continuous.
- $\nexists v_1 \neq v_2 : \kappa(v_1) = \kappa(v_2)$ , thus  $\kappa(v)$  is monotone.
- $\kappa(v) \rightarrow \infty$  as  $v \rightarrow v_{\min}$ .
- $\kappa(v) \rightarrow \kappa_{crit} \geq 0$  as  $v \rightarrow v_{\max}$ .
- When  $\kappa < \kappa_{crit}$  and  $v = v_{\max}$  there is a saddle to saddle-node travelling wave

$\kappa(v)$  is monotone and continuous thus there exists an inverse function satisfying the Theorem.

# Phase portrait, fixed points, isoclines

We consider travelling wave dynamical system:

$$\begin{aligned} A(s, c)s_\xi &= f(s, c) - v(s + d_1), \\ \kappa c_\xi &= v(d_1 c - d_2 - a(c)), \end{aligned}$$

Isoclines:

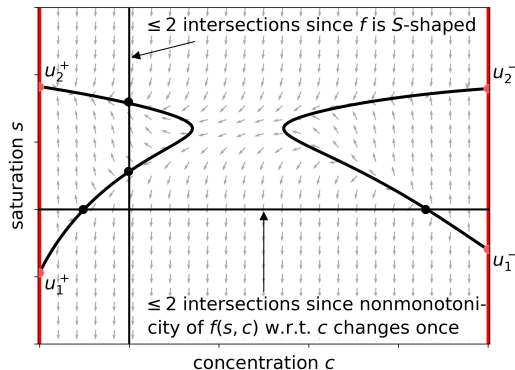
**red lines** are  $d_1 c - d_2 - a(c) = 0$ ,  
**black lines** are  $f(s, c) - v(s + d_1) = 0$ .

Fixed points:

$u_1^+$  and  $u_2^-$  — saddle points;  
 $u_2^+$  — attractor;  $u_1^-$  — repeller

Aim:

find all pairs  $(\kappa, v)$  for which there exists a trajectory  
 from saddle point  $u_2^-$  to saddle point  $u_1^+$



# Nullcline configurations: main and intermediate classes

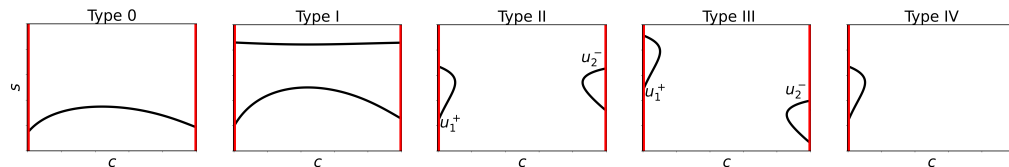


Figure 4: Five wide classes of nullcline configurations

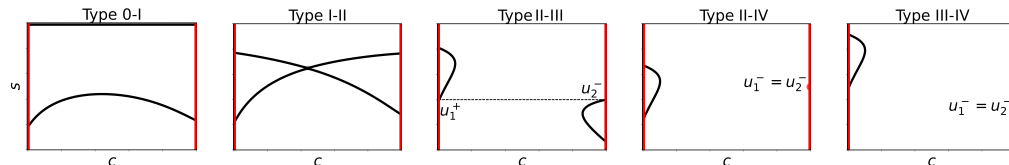


Figure 5: Intermediate types of nullcline configurations, appearing under assumpt. (F1)–(F4)

- Only Type II nullcline configuration has saddle-to-saddle connections.
- Type I-II corresponds to  $v_{\min}$ .
- Type II-III or Type II-IV correspond to  $v_{\max}$ .

# Nullcline configurations: monotone dependence on $v$

black lines  $f(s, c) = v(s + d_1)$

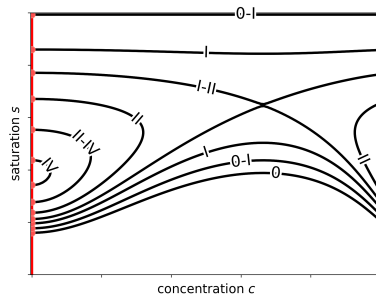
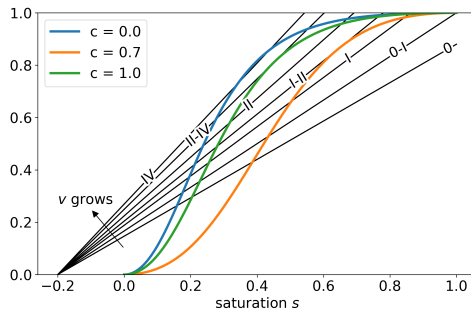


Figure 6: nullcline configuration evolution as  $v$  grows: Type 0  $\rightarrow$  Type I  $\rightarrow$  Type II  $\rightarrow$  Type IV

# Nullcline configurations: bad cases

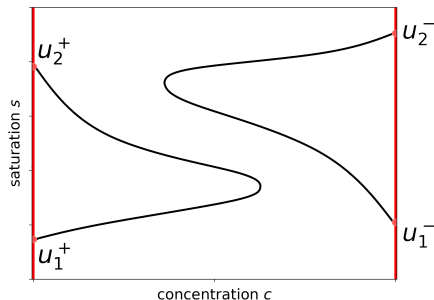


Figure 7: If  $f$  is not S-shaped.

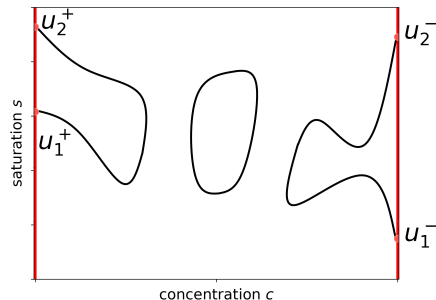


Figure 8: If non-monotonicity is more complex.

We believe that the similar result is true without conditions (F3)–(F4).

# Type II configuration: for every $v$ there exist $\kappa$

$$\frac{s_\xi}{c_\xi} = \kappa \cdot \frac{v^{-1}f(s, c) - (s + d_1)}{A(s, c)(d_1c - d_2 - a(c))}$$

Used property: continuous and monotonous dependence of trajectories on  $\kappa$ .

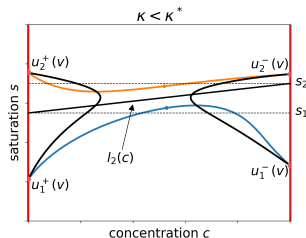


Figure 9:  $\kappa \ll 1$ .

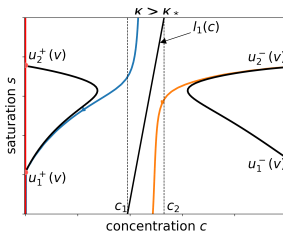


Figure 10:  $\kappa \gg 1$ .

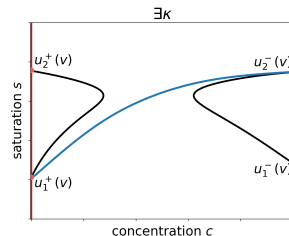


Figure 11:  $\exists \kappa$ .

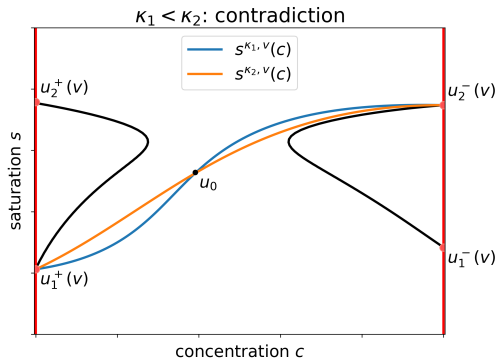
Type II config.:  $\kappa(v)$  is unique for every  $v \in (v_{\min}, v_{\max})$ .

If there are  $\kappa_1 < \kappa_2$  for one  $v$ , then the corresponding trajectories must intersect, which leads to a contradiction.

The slope

$$\frac{s_\xi}{c_\xi} = \kappa \cdot \frac{v^{-1}f(s, c) - (s + d_1)}{A(s, c)(d_1c - d_2 - a(c))}$$

is positive at the intersection point  $(s, c)$ , so it strictly increases when  $\kappa$  increases.



NB: this property might be lost for more complex nullcline configurations.

## Type II configuration: monotonicity of $\kappa(v)$

If  $\kappa(v_1) = \kappa(v_2)$  for  $v_1 < v_2$ , then the corresponding trajectories must intersect, which leads to a contradiction. The slope

$$\frac{s_\xi}{c_\xi} = \kappa \cdot \frac{v^{-1}f(s, c) - (s + d_1)}{A(s, c)(d_1c - d_2 - a(c))}$$

is positive at the intersection point  $(s, c)$ , so it strictly increases when  $v$  increases.

